

```

5.
6. Parameters:
7. ticker (str): Stock symbol
8. start_date (str): Start date in 'YYYY-MM-DD' format
9. end_date (str): End date in 'YYYY-MM-DD' format
10. param_bounds (dict): Dictionary with parameter names as keys and (min, max) tuples as
    values
11. population_size (int): Size of the parameter population in each generation
12. generations (int): Number of generations to evolve
13.
14. Returns:
15. pd.DataFrame: Results of the final generation's parameter combinations
16. """
17. import random
18.
19. # Prepare data once
20. data = prepare_backtest_data(ticker, start_date, end_date)
21.
22. # Function to evaluate a parameter set
23. def evaluate_params(short_ma, long_ma):
24.     # Calculate indicators
25.     data['SMA_Short'] = data['Adj Close'].rolling(window=short_ma).mean()
26.     data['SMA_Long'] = data['Adj Close'].rolling(window=long_ma).mean()
27.
28.     # Generate signals
29.     data['Signal'] = 0
30.     data.loc[data['SMA_Short'] > data['SMA_Long'], 'Signal'] = 1
31.
32. # Run backtest
33. backtest_result = backtest_strategy(data)
34.
35. # Evaluate results
36. metrics = evaluate_backtest(backtest_result)
37.
38. return metrics['Sharpe Ratio'] # Using Sharpe Ratio as fitness measure
39.

```